

The Common Stock Investment Performance of Individual Investors

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Abstract

Using account data for over 60,000 households from a large discount brokerage firm, we analyze the common stock investment performance of individual investors from February 1991 through December 1996. The average household tilts its common stock investment toward high-beta, small, value stocks, and turns over 80 per cent of its portfolio annually. On the one hand, the gross returns (before accounting for transaction costs) earned by the average household are unremarkable; the average household earned an annualized geometric mean gross return of 17.7 per cent while the value-weighted market index earned 17.1 per cent. On the other hand, the net returns earned by the average household lag reasonable benchmarks by economically and statistically significant amounts; the average household earned an annualized geometric mean net return of 15.3 per cent. The 20 per cent of households that trade most (averaging at least 9.6 per cent turnover per month) earned an annualized geometric mean net return of 10.0 per cent. We argue that the well-documented tendency for human beings to be overconfident can best explain the high trading levels and the resulting poor performance of individual investors. Our central message is that trading is hazardous to your wealth.

In 1996, approximately 47 per cent of equity investments in the U.S. were held directly by households, 23 per cent by pension funds, and 14 per cent by mutual funds (Security Industry Fact Book, 1997). Financial economists have extensively analyzed the return performance of equities managed by mutual funds. There is also a fair amount of research on the performance of equities managed by pension funds. Unfortunately, there is virtually no research on the return performance of equities held directly by households, despite their large ownership of equities.

In this paper, we attempt to shed light on the investment performance of common stocks held directly by households. To do so, we analyze a unique data set that consists of position statements and trading activity for 78,000 households at a large discount brokerage firm over a six-year period ending in December 1996. Our main descriptive results can be succinctly summarized. After a reasonable accounting for transaction costs, we document that these households significantly underperform relevant benchmarks. The households that we analyze earned gross returns (before accounting for transaction costs) that were remarkably close to those earned by an investment in a value-weighted index of NYSE/AMEX/NASDAQ stocks. During our sample period, an investment in a value-weighted market index earned an annualized geometric mean return of 17.1 per cent; the average household earned 17.7 per cent while in aggregate they earned 17.1 per cent. In contrast, the net performance (after accounting for the bid-ask spread and commissions) of these households was below par; the average household earned 15.3 per cent while in aggregate they earned 15.6 per cent. The empirical tests supporting these conclusions come from time-series regressions that employ either the Capital Asset Pricing Model (CAPM) or the three-factor model developed by Fama and French (1993) as our benchmark.

Though our descriptive results are intrinsically interesting on their own, our analyses also allow us to test two competing theories of trading activity. Using a rational expectation framework, Grossman and Stiglitz (1980) argue that investors will trade when the marginal benefit of doing so is equal to or exceeds the marginal cost of the trade. In contrast, Benos (1997), Odean (1998a), Daniel, Hirshleifer, and Subramanyam

(1998), Gervais and Odean (1998), and Caballé and Sákovics (1998) develop theoretical models of financial markets where investors suffer from overconfidence. The overconfidence models predict that investors will trade to their detriment.

Our most dramatic empirical evidence supports the view that overconfidence leads to excessive trading. On one hand, there is very little difference in the gross performance of households that trade frequently (with monthly turnover in excess of 9.6 per cent) and those who trade infrequently. In contrast, households that trade frequently earn a net annualized geometric mean return of 10.0 per cent, while those that trade infrequently earn 17.5 per cent. These results are consistent with models where trading emanates from investor overconfidence, but are inconsistent with models where trading results from rational expectations. Though liquidity, risk-based rebalancing, and taxes can explain some trading activity, we argue that it belies common sense that these motivations for trade, even in combination, can explain average annual turnover of almost 300 per cent for those households that trade most.

Our descriptive analysis also provides several additional conclusions that are noteworthy:

1. Households trade common stocks frequently. The average household turns over more than 80 per cent of its common stock portfolio annually.
2. Trading costs are high. The average round-trip trade cost 5 per cent in commissions and 1 per cent in bid-ask spread.
3. Households tilt their investments toward small, high-beta stocks. There is a less obvious tilt toward value (high book-to-market) stocks.

It is the cost of trading and the frequency of trading, not portfolio selection, that explain the poor investment performance of households during our sample period. In fact, their tilt toward small stocks, and to a lesser extent value stocks, helped their performance during our sample period (during which small stocks outperformed large stocks by 17 basis points per month and value outperformed growth by 24 basis points per month).¹

¹ These figures are based on the mean return from February 1991 through December 1996 for the size and book-to-market factors constructed by Fama and French (1993). In the remainder of this paper, when we refer to a size or value premium, our inference is based on the returns of these zero-investment portfolios.

The remainder of this paper is organized as follows. We discuss related research in section I. We discuss our data and empirical methods in section II. Our main descriptive results are presented in section III. We test the models of investor overconfidence in section IV. We discuss liquidity, risk, and taxes as motivations for trading in section V. Concluding remarks are made in section VI.

I. Related Research

To our knowledge, the current investigation is the first comprehensive study of the aggregate common stock performance of individual investors who manage their own equity investments. However, there are two related studies that analyze the profitability of common stock trades (as distinct from positions held) by individual investors.

Schlarbaum, Lewellen, and Lease (1978) analyze the gross and net returns of round-trip trades made by 2,500 accounts at a retail brokerage firm over a seven-year period ending in December 1970. Though they emphasize that their results are conjectural, they conclude (p.322) that their results "portray an overall picture of quite respectable individual investor security selection acumen." In contrast, we document that individual investors at a discount brokerage firm during the six-year period ending December 1996 performed poorly.

There are at least three reasons why our results might differ from those in Schlarbaum et al. First, we analyze households that hold their investments at a discount brokerage, rather than a retail brokerage, firm. If retail brokerage firms provide valuable stock selection advice, it is plausible that individual investors at these firms earn both better gross returns and net returns. If investors at retail brokerage firms trade less frequently, it is likely that they earn better net returns. We would welcome the opportunity to test these hypotheses directly by obtaining a data set similar to that employed in our study from a retail brokerage firm.

Second, the analysis in Schlarbaum et al. focused on the returns from round-trip trades. There is now evidence that investors have a tendency to sell winning investments and hold on to losing investments (Odean, 1998b). Thus by analyzing trades, rather than position statements (as we do in the current study), Schlarbaum et al. may upwardly bias their return estimates.

Third, Schlarbaum et al. evaluate performance using a variety of market indexes. However, they do not explicitly adjust for any components of risk. In both our work and theirs, individual investors tilt toward high-beta stocks. At least in our study, adjusting for this higher market risk can affect the conclusions that one reaches. More importantly, however, is the tendency for individual investors to tilt toward small stocks. Though Schlarbaum et al. do not explicitly address whether such a tilt exists in their trade data, we suspect that it does. Schlarbaum et al. do not perform any adjustments for the tilt toward small stocks (though of course firm size did not have the same celebrity status in 1978 that it enjoys today). This is likely extremely important, since small stocks outperformed large stocks by 67 basis points *per month* during their sample period.

As do Schlarbaum et al., Odean (1998c) also focuses on the trades of individual investors. He analyzes the timing of trades made by individual investors at a large discount brokerage firm during the 7 years ending in December 1993, a sample period that overlaps with ours. (The data sets employed in Odean (1998c) and this study are different.) He documents that the stocks that individuals sell subsequently outperform the stocks that they buy. Thus, the implications of his study and the current investigation are similar: individual investors trade too much. However, Odean (1998c) does not analyze the aggregate performance of all stocks held by individuals. Consequently, he is unable to conclude whether individual investors perform well or not in aggregate, the focus of our investigation.

II. Data and Methods

A. Household Account Data

Our primary data set for this research is information from a large discount brokerage firm on the investments of 78,000 households from February 1991 through December 1996. The data set includes all accounts opened by each household at this discount brokerage firm. The sample selection was performed at the household level and was stratified based on whether the discount brokerage firm labeled the household as a general (60,000 households), affluent (12,000 households), or active trader household (6,000 households). The firm labels households that make more than 48 trades in any year as active traders, households with more than \$100,000 in equity at any point in time as affluent, and all other households as general. If a household qualifies as either active trader or affluent, it is assigned the active trader label. Sampled households were required to have an open account with the discount brokerage firm during 1991. Roughly half of the accounts in our analysis were opened prior to 1987, while half were opened between 1987 and 1991.

In this research, we focus on the common stock investments of households. We exclude from the current analysis investments in mutual funds (both open- and closed-end), American depository receipts (ADRs), warrants, and options. In Table 1, we present descriptive information on our sample. Of the 78,000 sampled households, 64,715 had positions in common stocks during at least one month; the remaining accounts either held cash or investments in other than individual common stocks. The average household had approximately two accounts, and roughly 60 per cent of the market value in these accounts was held in common stocks. There were over 3 million trades in all securities; common stocks accounted for slightly more than 60 per cent of all trades. The average household held 4 stocks worth \$41,344 during our sample period, though each of these figures is positively skewed. The median household held 2.48 stocks worth \$14,579. In aggregate, these households held more than \$2.5 billion in common stocks during our 71-month sample period, on average.

In Table 2, we present descriptive information on the trading activity for our sample. Panel A presents information on purchases, while Panel B contains information on sales. There were slightly more purchases (1,082,107) than sales (887,594) during our sample period, though the average value of stocks sold (\$13,707) was slightly higher than the value of stocks purchased (\$11,205). As a result, the aggregate value of purchases and sales were roughly equal (\$12.1 and \$12.2 billion, respectively). The average trade was transacted at a price of \$31 per share. The value of trades and the transaction price of trades are positively skewed; the medians for both purchases and sales are substantially less than the mean values.

For each trade, we estimate the bid-ask spread component of transaction costs for purchases (spr_{d_b}) or sales (spr_{d_s}) as:

$$spr_{d_s} = \frac{P_{d_s}^{cl}}{P_{d_s}^s} - 1, \text{ and}$$

$$spr_{d_b} = -\frac{P_{d_b}^{cl}}{P_{d_b}^b} - 1.$$

$P_{d_s}^{cl}$ and $P_{d_b}^{cl}$ are the reported closing prices from the Center for Research in Security Prices (CRSP) daily stock return files on the day of a sale and purchase, respectively; $P_{d_s}^s$ and $P_{d_b}^b$ are the actual sale and purchase price from our account database. Our estimate of the bid-ask spread component of transaction costs includes any market impact that might result from a trade. It also includes an intraday return on the day of the trade. (In Appendix A, we provide a detailed reconciliation of our return calculations.) The commission component of transaction costs is estimated as the dollar value of the commission paid scaled by the total principal value of the transaction, both of which are reported in our account data.

The average purchase cost an investor 0.31 per cent in bid-ask spread and 2.09 per cent in commissions, though the median purchase incurred no spread costs and 1.40 per cent in commissions. The average sale cost an investor 0.69 per cent in bid-ask spread and 3.07 per cent in commissions, though the median sale incurred virtually no spread costs and 1.25 per cent in commissions. Our estimate of the bid-ask spread is very close

to the trading cost of 0.21 per cent for purchases and 0.63 per cent for sales paid by open-end mutual funds from 1966 to 1993 (Carhart, 1997).

In Panels C and D of Table 2, we calculate the trade-weighted (weighted by trade size) spreads and commissions. These figures can be thought of as the total cost of conducting the \$24 billion in common stock trades (\$12 billion each in purchases and sales). Trade size has little effect on spread costs (0.27 per cent for purchases and 0.69 per cent for sales) but substantially reduces the commission costs (0.77 per cent for purchases and 0.66 per cent for sales).

In sum, the average trade incurred a round-trip transaction cost of about 1 per cent for the bid-ask spread and about 5 per cent in commissions. In aggregate, round-trip trades cost about 1 per cent for the bid-ask spread and about 1.4 per cent in commissions.

Finally, we calculate the monthly portfolio turnover for each household. In each month during our sample period, we identify the common stocks held by each household at the beginning of month t from their position statement. To calculate monthly sales turnover, we match these positions to sales during month t . The monthly sales turnover is calculated as the shares sold times the beginning-of-month price per share divided by the total beginning-of-month market value of the household's portfolio. To calculate monthly purchase turnover, we match these positions to purchases during month $t-1$. The monthly purchase turnover is calculated as the shares purchased times the beginning-of-month price per share divided by the total beginning-of-month market value of the portfolio.² The average household purchased 7.28 per cent and sold 6.58 per cent of their stock portfolio each month, though the median household traded much less frequently (buying 2.95 per cent of their stock portfolio, while selling 2.48 per cent). In panels C and D, we calculate trade-weighted purchase (sales) turnover by summing all purchases (sales) and dividing by the sum of all positions during our sample period. The trade-weighted

² If more shares were sold than were held at the beginning of the month (because, for example, an investor purchased additional shares after the beginning of the month), we assume the entire beginning-of-month position in that security was sold. Similarly, if more shares were purchased in the preceding month than

purchase turnover was 4.91 per cent, while the trade-weighted sales turnover was 4.47 per cent.

In sum, these investors traded its common stocks quite frequently. The average household turned over more than 80 per cent of its common stock portfolio each year. This is uncannily close to the average turnover reported by U.S. common stock mutual funds from 1966 to 1993 of 77 per cent (Carhart, 1997). In aggregate, these investors turned over more than 50 per cent of their invested wealth each year.

B. Measuring Return Performance

The focus of our analysis is the return performance of investments in common stocks by households. We analyze both the gross performance and net performance (after a reasonable accounting for commissions, the bid-ask spread, and the market impact of trades).

We estimate the gross monthly return on each common stock investment using the beginning-of-month position statements from our household data and the CRSP monthly returns file. In so doing, we make two simplifying assumptions. First, we assume that all securities are bought or sold on the last day of the month. Thus, we ignore the returns earned on stocks purchased from the purchase date to the end of the month and include the returns earned on stocks sold from the sale date to the end of the month. Second, we ignore intramonth trading (e.g., a purchase on March 6 and a sale of the same security on March 20), though we do include in our analysis short-term trades that yield a position at the end of a calendar month.

In appendix A, we document that accounting for the exact timing of trades would reduce the performance of individual investors by about two basis points per month. In appendix B, we document that accounting for intramonth trades would improve the performance of individual investors reported in our main results by about one basis point per month. More importantly, a careful accounting for both the exact timing of trades

were held in the position statement, we assume the entire position was purchased in the preceding month. Thus, turnover, as we have calculated it, cannot exceed 100 per cent in a month.

and the profitability of intramonth trades indicates the results that we report in the main text are slightly high for our full sample and every sample partition that we analyze.

Consider the common stock portfolio for a particular household. The gross monthly return on the household's portfolio (R_{ht}^{gr}) is calculated as:

$$R_{ht}^{gr} = \sum_{i=1}^{s_{ht}} p_{it} R_{it}^{gr},$$

where p_{it} is the beginning-of-month market value for the holding of stock i by household h in month t divided by the beginning-of-month market value of all stocks held by household h , R_{it}^{gr} is the gross monthly return for that stock, and s_{ht} are the number of stocks held by household h in month t .

For security i in month t , we calculate a monthly return net of transaction costs (R_{it}^{net}) as:

$$(1 + R_{it}^{net}) = (1 + R_{it}^{gr}) \frac{(1 - c_{it}^s)}{(1 + c_{i,t-1}^b)}$$

where c_{it}^s is the cost of sales scaled by the sales price in month t and $c_{i,t-1}^b$ is the cost of purchases scaled by the purchase price in month $t-1$. The cost of purchases and sales include the commissions and bid-ask spread components, which are estimated individually for each trade as previously described. Thus, for a security purchased in month $t-1$ and sold in month t , both c_{it}^s and $c_{i,t-1}^b$ are positive; for a security that was neither purchased in month $t-1$ nor sold in month t , both c_{it}^s and $c_{i,t-1}^b$ are zero. Because the timing and cost of purchases and sales vary across households, the net return for security i in month t will vary across households. The net monthly portfolio return for each household is:

$$R_{ht}^{net} = \sum_{i=1}^{s_{ht}} p_{it} R_{it}^{net}.$$

(If only a portion of the beginning-of-month position in stock i was purchased or sold, the transaction cost is only applied to the portion that was purchased or sold.)

We estimate the aggregate gross and net monthly return earned by individual investors as:

$$RAG_t^{gr} = \sum_{h=1}^{n_{ht}} x_{ht} R_{ht}^{gr}, \text{ and}$$

$$RAG_t^{net} = \sum_{h=1}^{n_{ht}} x_{ht} R_{ht}^{net},$$

where n_{ht} are the number of households with common stock investment in month t and x_{ht} is the beginning-of-month market value of common stocks held by household h divided by the beginning-of-month market value of common stock held by all households. We estimate the gross and net monthly return earned by the average household as:

$$RH_t^{gr} = \frac{1}{n_{ht}} \sum_{h=1}^{n_{ht}} R_{ht}^{gr}, \text{ and}$$

$$RH_t^{net} = \frac{1}{n_{ht}} \sum_{h=1}^{n_{ht}} R_{ht}^{net}.$$

III. Results

A. Full Sample Results

Our main findings for the full sample can be summarized simply. The gross returns earned by individual investors in aggregate (RAG_t^{gr}) and the gross return earned by the average household (RH_t^{gr}) are remarkably close to that earned by an investment in a value-weighted index of NYSE/AMEX/NASDAQ stocks.³ The annualized geometric mean return earned by individual investors in aggregate, the average household, and the value-weighted market index are 17.1, 17.7, and 17.1 per cent, respectively. In contrast, the net returns earned by individual investors in aggregate (RAG_t^{net}) and the net return earned by the average household (RH_t^{net}) underperform the value-weighted index by

³ We use the NYSE/AMEX/NASDAQ value-weighted market index constructed by Fama and French (1993). Firms comprising the index must have data for firm size and book-to-market ratio. The correlation between this market index and the NYSE/AMEX/NASDAQ value-weighted index from CRSP is very high.

more than 100 basis points annually. The net annualized geometric mean return earned by individual investors in aggregate and the average household are 15.6 and 15.3 per cent, respectively.

Of course, more refined conclusions require a thorough investigation of risk-adjusted performance. To address this issue, we calculate three measures of risk-adjusted performance. First, we calculate the mean monthly market-adjusted abnormal return for individual investors by subtracting the return on a value-weighted index of NYSE/ASE/NASDAQ stocks from the return earned by individual investors.

Second, we employ the theoretical framework of the Capital Asset Pricing Model and estimate Jensen's alpha by regressing the monthly excess return earned by individual investors on the market excess return. For example, to evaluate the gross monthly return earned by individual investors in aggregate, we estimate the following monthly time-series regression:

$$RAG_t^{gr} - R_{ft} = a_i + b_i(R_{mt} - R_{ft}) + e_i$$

where:

R_{ft} = the monthly return on T-Bills,⁴

R_{mt} = the monthly return on a value-weighted market index,

a_i = the estimated CAPM intercept (Jensen's alpha),

b_i = the estimated market beta, and

e_i = the regression error term.

The subscript i denotes parameter estimates and error terms from regression i , where we estimate four regressions: one each for the gross and net performance of individual investors in aggregate, and one each for the gross and net performance of the average household.

⁴ The return on T-bills is from Stocks, Bonds, Bills, and Inflation, 1997 Yearbook, Ibbotson Associates, Chicago, IL.

Third, we employ an intercept test using the three-factor model developed by Fama and French (1993). For example, to evaluate the performance of individual investors in aggregate, we estimate the following monthly time-series regression:

$$RAG_t^{gr} - R_{ft} = a_j + b_j(R_{mt} - R_{ft}) + s_jSMB_t + h_jHML_t + e_j$$

where SMB_t is the difference in the returns of a value-weighted portfolio of small stocks and big stocks and HML_t is the difference in returns of a value-weighted portfolio of high book-to-market stocks and low book-to-market stocks.⁵ The regression yields parameter estimates of a_j , b_j , s_j , and h_j . The error term in the regression is denoted by e_j . The subscript j denotes parameter estimates and error terms from regression j , where we again estimate four regressions. We place particular emphasis on the Fama-French intercept tests, since individual investors tilt their portfolios toward small stocks. The three-factor model provides a reasonable adjustment for this small stock tilt.⁶

The results of this analysis are presented in Table 3. Panel A present results for the gross performance of individual investors in aggregate, while panel B presents results for the average household. The three performance measures indicate that the gross performance of individual investors is unremarkable; neither the market-adjusted return, Jensen's alpha, nor the intercept test from the Fama-French three-factor model are reliably different from zero.

Also noteworthy in these results are the coefficient estimates on the market, size, and book-to-market factors. Individual investors tilt toward small stocks with high market risk. The market beta for stocks held by individual investors is reliably greater than one and the coefficient estimate on SMB_t is reliably positive. Though in aggregate, individual investors have no tilt toward value or growth, the average household has a slight tilt toward value stocks (those with high book-to-market ratios) and a more

⁵ The construction of these portfolios is discussed in detail in Fama and French (1993). We thank Kenneth French for providing us with these data.

pronounced tilt toward small stocks. These tilts served individual investors well during our period of analysis; the mean monthly returns on SMB_t and HML_t during our 71-month sample period were 0.17 and 0.24 per cent, respectively. This observation can account for the fact that the market-adjusted return performance of individual investors is positive (albeit unreliably so), while Jensen's alpha (CAPM intercept) and the intercept test from the Fama-French three-factor model are negative.

The more interesting findings of our analysis are contained in Panels C and D. Net of transaction costs, individual investors perform poorly. Individual investors underperform a value-weighted market index by about 10 or 11 basis points per month, though this underperformance is not reliably different from zero. Of course, a simple market-adjustment does not account for the tendency for individual investors to tilt toward small stocks with high market risk (and, to a lesser extent, value stocks). Jensen's alpha (CAPM intercept) indicates individual investors underperform by 18 or 19 basis points per month. The intercept tests from the Fama-French three-factor model indicate aggregate underperformance of 20 basis points per month, while the average household underperforms by 33 basis points per month. In combination, these results indicate that the net return performance of individual investors was reliably negative.

B. Sorting on Portfolio Size and Household Type

To test the robustness of our results, we conduct two auxiliary analyses. First, we present results based on the three different household types represented in our sample: general, affluent, and active trader. Descriptive statistics on the partition by household type are presented in Table 4. Predictably, general households held much less common stock than did households labeled as affluent or active trader. The active trader households have much higher monthly turnover rates (12.78 per cent) than do general or affluent households (6.74 and 4.60 per cent respectively). Here and in the remainder of the paper when we report turnover, we are reporting the average of purchase and sale turnover.

⁶ Lyon, Barber, and Tsai (1998) document that intercept tests using the three-factor model are well specified in random samples and samples of large or small firms. Thus, the Fama-French intercept tests employed here account well for the small stock tilt of individual investors.

As before, we estimate the parameters of the Fama-French three-factor model, where the dependent variable is the monthly mean gross household excess return for each household type. (In the interest of parsimony, here and in the remainder of the paper we do not report results for the aggregate performance of each partition. We note when conclusions are different using the aggregate performance.) The coefficient estimates on the market, size, and book-to-market factors reveal that general and active trader households tilt more heavily toward small stocks than do affluent households. There is also some evidence that the active trader households bear more market risk than do general or affluent household types; the active trader households also tilt more toward growth stocks than do general or affluent households.

The gross and net performance for each of the three household types are presented in Table 5. Inspection of Panel A reveals that none of the household types have superior gross return performance. Though all household types have positive gross market-adjusted returns, none are reliably different from zero (and, of course, the market-adjustment does not account for the tilt toward small stocks). The Fama-French intercept test provides some evidence that the gross performance of general households is poor. Panel B reveals that the net performance for each of the household types is different. General and active trader households significantly underperform, while the affluent households have no unusual performance. Though the market-adjusted and CAPM intercept tests are not reliably different from zero, these tests do not account for the tilt toward small stocks.⁷ The Fama-French intercept tests, which account for the small stock tilt, document economically and statistically significant underperformance for the general and active trader households, but not for the affluent households.

The dominant performance of the affluent households is easily explained. These households trade less. (Remember that wealthy households that trade actively are categorized as active traders rather than affluent.) Furthermore, because they have larger

positions in common stocks, the proportional cost of commissions for these households is less than for general households. While the active traders would also benefit from lower proportional commission costs (since on average the value of the common stocks they hold is greater than that of the affluent households), they trade at almost three times the rate of affluent households.

In our second auxiliary analysis, we test the robustness of our results across different position sizes by partitioning the households into quintiles on the basis of portfolio size. We define portfolio size as the market value of common stocks held in the first month for which there is a position statement. (If the first position statement appears after January 1991, we do not discount the market value of the common stocks to January 1991 in our rankings. Our results are virtually identical if we discount the market value of these common stocks using the return on the value-weighted market index.) Each quintile represents the common stock investments of more than 12,000 households.

Descriptive statistics on the partition by portfolio size are presented in Table 6. The largest portfolios have a mean beginning position market value of \$133,262, while the smallest portfolios average \$1,599. Small portfolios have slightly higher monthly turnover (7.44 per cent) than large portfolios (6.72 per cent). As before, we estimate the parameters of the Fama-French three-factor model, where the dependent variable is the monthly mean gross household excess return for each quintile. The coefficient estimates on the market, size, and book-to-market factors reveal that small portfolios tilt more heavily towards high-beta, small, value stocks than do large portfolios.

The gross and net performance for each quintile are presented in Table 7. Focusing first on the gross performance (Panel A), we find that small portfolios (quintile 1) earn higher average returns than large portfolios (quintile 5), though the difference is not reliably different from zero. This difference is likely attributable to the fact that small portfolios tilt more heavily toward small value stocks, which performed well during our

⁷ The aggregate performance of general accounts is reliably negative using Jensen's alpha (-0.336 per cent, $t=-2.57$), though that of the active traders (-0.223 per cent, $t=-1.45$) and affluent accounts (-0.029 per

sample period. The net performance results are presented in Panel B. The market-adjusted return and Jensen's alpha are similar to those reported for the full sample for each quintile. Though the point estimates are consistently negative, they are not reliably so. Of course, these risk-adjustments ignore the fact that investors are tilting towards small value stocks. In contrast, the intercept tests from the Fama-French three-factor model indicate significant underperformance, ranging from 27 to 40 basis points per month, in each of the quintiles. In sum, after a reasonable accounting for the size and value tilts of small investors, we document that both small and large portfolios underperform.

C. *Cross-Sectional Variation in Performance*

We should emphasize that the aggregate performance and average household performance, though germane and interesting, mask considerable cross-sectional variation in the performance across households. For each household, we calculate the mean monthly market-adjusted abnormal return. We present the distribution of these means in Table 8 and Figure 1. (We omit from this analysis accounts that held common stocks for less than 12 months during our 71-month sample period.) Consistent with the results presented in Table 3, the median household earned a gross monthly market-adjusted return of 0.04 per cent and a net return of -0.09 per cent. Though 51.8 per cent of households outperformed a value-weighted market index before transaction costs, only 46 per cent outperformed the index after costs. Nonetheless, many household perform very well; 25 per cent of all households beat the market, after accounting for transaction costs by more than 0.63 per cent per month (more than 7 per cent annually). Conversely, many households perform very poorly; 25 percent of all households underperform the market, after accounting for transaction costs, by more than 0.77 per cent per month (more than 9 per cent annually).

IV. Overconfidence and Performance

It is well documented that people tend to be overconfident (e.g. Alpert and Raiffa (1982), Griffen and Tversky (1992); see Odean (1998a) for a more detailed review).

cent, $t=-0.46$) is not.

Benos (1997), Odean (1998a), Daniel, Hirshleifer, and Subramanyam (1998), Gervais and Odean (1998), and Caballé and Sákovics (1998) develop theoretical models of financial markets where investors suffer from overconfidence. These models yield one general prediction: overconfident investors will trade too much (i.e. trading, at the margin, will reduce their expected utility).⁸ In contrast, in a rational expectation framework, Grossman and Stiglitz (1980) argue that investors will trade when the marginal benefit of doing so is equal to or exceeds the marginal cost of the trade (including the cost of acquiring information). Odean (1998a) analyzes a variation of Grossman and Stiglitz's model in which investors are overconfident. The two models yield different predictions about the gains of trading. The rational model predicts that investors who trade more (i.e. those whose expected trading is greater) will have the same expected utility as those who trade less. The overconfidence model predicts that investors who trade more will have lower expected utility.

Consider the implications of these two models in our empirical setting. The overconfidence model predicts that the net return performance of households with high turnover will be lower than that of households with low turnover, while making no prediction about the differences in gross returns. The Grossman-Stiglitz model predicts that the gross return performance of households with high turnover will be higher than that of households with low turnover, but there will be little difference in the net returns.

To test these competing models, we partition our sample of households into quintiles on the basis of mean monthly turnover (defined as the average of purchase and sale turnover). Each quintile represents the common stock investments of more than 12,000 households. Descriptive statistics for each of the quintiles are presented in Table 9. The households with low turnover average 0.12 per cent turnover per month, while those with high turnover average 23.62 per cent. To make it into the high turnover portfolio, a household would need to turn over at least 9.6 per cent of their portfolio in an average month.

⁸ In an exception to this finding, Kyle and Wang (1997) argue that when traders compete for duopoly profits, overconfident traders may reap greater profits. This prediction is based on several assumptions

As before, we estimate the parameters of the Fama-French three-factor model, where the dependent variable is the monthly mean gross household excess return for each turnover quintile. The coefficient estimates on the market, size, and book-to-market factors reveal that the high turnover households tilt more heavily towards high-beta, small, growth stocks than do the low turnover households.

The gross and net performance for each turnover quintile are presented in Table 10. Focusing first on the gross performance (Panel A), we find that high turnover (quintile 5) households do not significantly outperform low turnover households (quintile 1). In fact, the intercept test based on the Fama-French three-factor model, which accounts for the tendency for the high turnover portfolio to tilt toward high-beta, small, growth stocks, indicates that the two high turnover quintiles (quintiles 4 and 5) underperform by 27 and 37 basis points per month. Though marginally statistically significant, we believe these figures to be economically large (approximately 3 to 4 per cent annually). Regardless of whether one accepts these results as statistically significant, the prediction of the Grossman and Stiglitz model is not supported; those who trade most do not earn higher gross returns.

The analysis of net returns (Panel B) is quite interesting. Regardless of the method used to measure performance, the high turnover households (quintile 5) underperform the low turnover households (quintile 1). The intercept tests from the Fama-French three-factor model indicate this underperformance is an astoundingly high 91 basis points per month. Again, these differences are not consistent with the Grossman and Stiglitz model, but are consistent with the predictions of the overconfidence models.

In sum, differences in gross returns across the turnover quintiles are small. An investment that mimicked that of the average household in each quintile would have earned a gross annualized mean geometric return that ranged between 17.4 per cent (for quintile 4) to 17.8 per cent (for quintile 2). However, there are dramatic differences in

that do not apply to individuals trading common stocks.

the net returns across the turnover quintiles. An investment that mimicked the average household of the high turnover quintile would have earned a net annualized mean geometric return of 10.0 per cent, while an investment that mimicked the low turnover quintile would have earned 17.5 per cent.

V. Liquidity, Rebalancing, and Tax-Motivated Trading

To this point, we have focused on information-motivated versus overconfidence-motivated trading. The empirical evidence that we have presented solidly favors overconfidence as the major motivation for trading, since trading unambiguously hurts investor performance. But, there are other motivations for trading, which we consider in this section.

A. Liquidity

Investors who face liquidity shocks over time will trade as a rational response to those shocks. Thus, liquidity shocks can explain some trading activity. However, liquidity shocks as an explanation of the 80 per cent annual turnover that we document for the average individual investor seems implausible; liquidity shocks as an explanation of the almost 300 per cent annual turnover of the households who trade most belies common sense.

Moreover, the trading that results from liquidity shocks can be accomplished at a much lower cost by investing in mutual funds than by investing in individual common stocks. To illustrate this point, we analyze the returns on the Vanguard Index 500 mutual fund, a large passive mutual fund that claims to match the performance of the Standard and Poor's 500. Investors can move in and out of this fund at no cost.

Table 11 presents the results of this analysis. In contrast to the performance of the average or aggregate household, this index fund does not underperform using any of the standard performance benchmarks. During our sample period, this fund earned an annualized geometric mean return of 16.9 per cent while the value-weighted market index earned 17.1 per cent. A passively managed mutual fund clearly provides a lower

cost means of managing liquidity shocks than does investment in individual common stocks.

B. *Rebalancing*

Investors who desire a portfolio with certain risk characteristics will rationally rebalance their portfolio to maintain this risk profile. We believe that risk-based rebalancing is not a significant motivation for trading in the households that we study. Risk-based rebalancing as an explanation of the 80 per cent annual turnover that we document for the average household belies common sense. In addition, investors can manage the risk composition of their portfolio at much lower cost by carefully selecting a portfolio of mutual funds.

C. *Taxes*

The single most compelling reason for investors to hold individual common stocks in lieu of mutual funds is taxes. Investors who hold stocks that have lost value since their purchase can realize those losses. These losses can be used to shelter gains and thereby reduce the investor's tax liability.⁹

Tax-loss selling cannot completely explain the results that we document here for three reasons. First, it is implausible that tax-motivated trading would yield an annual turnover rate of 80 per cent.

Second, Odean (1998b, 1998c) documents that most investor trading activity is inconsistent with tax-motivated trading. He observes that investors at a discount brokerage sell profitable investments twice as often as unprofitable investments (during the period 1987-1993) and that, relative to their opportunities to do so, these investors are about one and a half times more likely to realize any gain than any loss. They do engage in tax-loss selling late in the year, but December is the only month in which they realize losses at as fast a rate as they do gains.

⁹ Though losses on mutual funds can also be used to reduce an investor's tax liability, the probability of having a loss on a mutual fund is less than the probability of observing at least one losing investment in a well-diversified portfolio of common stocks.

Third, we find significant underperformance in both taxable and tax-deferred accounts. If tax-loss selling is the major motivation for trading, we would not expect to find significant underperformance in the tax-deferred accounts. We partition the accounts in our sample into taxable and tax-deferred accounts (i.e. Individual Retirement Accounts and Keogh Accounts). In Table 12, we present descriptive statistics for the taxable and tax-deferred accounts. The taxable accounts are larger and have a stronger tilt toward small growth firms than do the tax-deferred accounts. The taxable accounts also have higher monthly turnover (8 per cent) than the tax-deferred accounts (6 per cent). This difference in turnover may result from tax-motivated trading or it may be that investors associate their retirement accounts with future safety and therefore trade less speculatively in these accounts.

In Table 13, we present the gross and net return performance of taxable and tax-deferred accounts. The gross returns earned by taxable and tax-deferred accounts are quite similar (see Panel A). The net returns earned by taxable and tax-deferred accounts are both poor, after a reasonable accounting for the small stock tilt of the individual investors that we analyze. The tax-deferred accounts outperform the taxable accounts by about six basis points per month. In short, the general tenor of our results is similar for both taxable and tax-deferred accounts.

VI. Conclusion

We analyze the returns earned on common stock investments by over 60,000 households at a large discount brokerage firm for the six years ending in December 1996. We document that the gross returns (before accounting for transaction costs) earned by these households are quite ordinary, on average. Unfortunately, the net returns (after accounting for the bid-ask spread and commissions paid by these investors) earned by these households are poor. The average household underperforms a value-weighted market index by about 10 basis points per month (or 1.2 per cent annually). After accounting for the fact that the average household tilts its common stock investments towards small value stocks with high market risk, the underperformance averages 33

basis points per month (or 4.0 per cent annually). The average household turns over about 80 per cent of its common stock portfolio annually. The poor performance of the average household can be traced to the costs associated with this high level of trading.

Our most dramatic empirical evidence is provided by the 20 per cent of households who trade most often. With monthly turnover of almost 25 per cent, on average, these households turn its common stock portfolios over almost three times annually. Though the gross returns earned by these high-turnover households are unremarkable, their net returns are anemic. The net returns lag a value-weighted market index by 46 basis points per month (or 5.5 per cent annually). After a reasonable accounting for the fact that the average high-turnover household tilts its common stock investments towards small value stocks with high market risk, the underperformance averages 91 basis points per month (or 10.9 per cent annually).

The investment experience of individual investors is remarkably similar to the investment experience of mutual funds. As do individual investors, the average mutual fund underperforms a simple market index (Jensen, 1969, and Malkiel, 1995). Mutual funds trade often and their trading hurts performance (Carhart, 1997). But trading by individual investors is even more deleterious to performance, since individuals execute small trades and face higher proportional commission costs than mutual funds.

Our main point is simple: trading is hazardous to your wealth. Why then do investors trade so often?. The aggregate turnover of the individual investor portfolios we analyze is about 50 per cent; the average turnover is about 80 per cent. The New York Stock Exchange reports that annual turnover of stocks listed on the exchange hovered around 50 per cent during our sample period. Mutual funds average annual turnover of 77 per cent (Carhart, 1997). We believe that these high levels of trading can be at least partially explained by a simple behavioral bias: People are overconfident; overconfidence leads to too much trading.

Based on rational agents free from such behavioral biases, the efficient markets hypothesis has been central to both the theory and practice of investment management. The efficiency research posits that private information is rare. Thus, active investment strategies will not outperform passive investment strategies. Both the theoretical and empirical work on efficiency supporting this view have led to a rise of passive investment strategies that simply buy and hold diversified portfolios (Fama (1991)). The models of investor overconfidence (e.g. Odean, 1998a) provide an even stronger prediction: active investment strategies will underperform passive investment strategies. Overconfident investors will overestimate the value of their private information. This will cause them to trade too actively and, consequently, to earn below average returns. Consistent with these models of investor overconfidence, we provide empirical evidence that households, which hold about half of U.S. equities, trade too much, on average. Those who trade the most are hurt the most.

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Appendix A: The Analysis of Trade Timing

In this appendix, we analyze the timing of purchases and sales within a month. The timing of trades within a month are ignored in our main analysis, since we assume all purchases and sales are made at month end. Consistent with the results reported in Odean (1998c), we document that the stocks investors buy underperform the stocks that they sell. In aggregate, we estimate that an exact accounting for the timing of purchases and sales would reduce the performance of individual investors by more than two basis points per month (or approximately 0.29 per cent annually).

For each account with a beginning-of-month position statement in month t , we identify all purchases in month $t-1$ and sales in month t . For both purchases and sales, we calculate the compound return on the stock from the day following the trade to the last day of the month. For purchases this return is excluded in our main results, while for sales this return is included. Note that in our main results, we account for the intraday return on the trade day in our estimate of the bid-ask spread.

The results of our analysis are presented in table A1. The second (fourth) column of this table present aggregate purchase (sale) turnover calculated as the aggregate dollar value of purchases (sales) divided by the aggregate dollar value of positions held. (This turnover measure is slightly different than that used in the main text, where turnover is calculated based on market values contained in position statements and is thus capped at 100 per cent per month for each household.) Abnormal returns are calculated for purchases and sales by subtracting the compound return on the CRSP NYSE/AMEX/NASDAQ value-weighted index. The trade-weighted mean abnormal returns are presented in columns three (for purchases) and five (for sales) of table A1. In aggregate (Panel A), from the day following the trade to the end of the month, the stocks that investors bought underperformed the value-weighted market index by 45 basis points, while those that they sold outperformed the index by one basis point. Based on these abnormal returns and our estimates of aggregate turnover, we calculate that the results we present in the main text overestimate the performance of individual investors by 2.45 basis points per month.

We also analyze the timing of trades by partitioning households on the basis of account size (Panel B), household type (Panel C), and turnover (Panel D). In each of the sample partitions, the timing of their trades hurts investors. In short, the results in the main text overestimate the performance of individual investors by ignoring the exact timing of purchases and sales.

Consider how the accounting for the exact timing of trades relates to the return calculations contained in the main text. In figure A1, we present an example of a security that is purchased in month one and sold in month three. A time line for these transactions is depicted in figure A1:

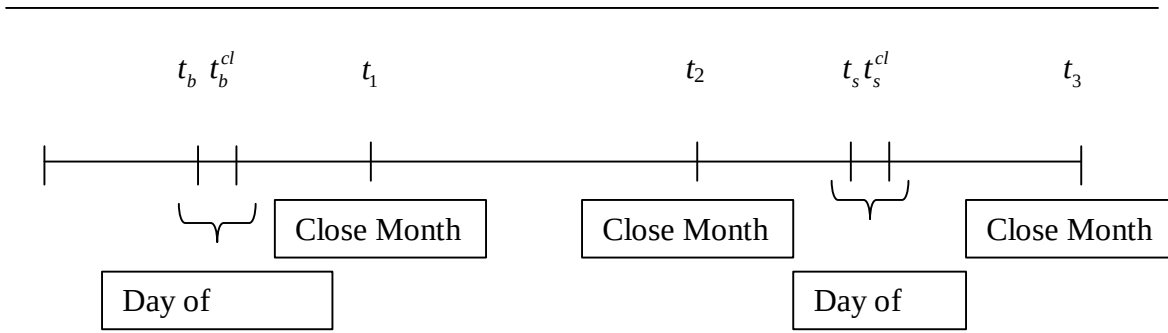


Figure A1: Time Line of Returns Calculations

In the main text, we calculate the return for this security from t_1 to t_3 . In this appendix, we calculate the return from timing as the return from t_b^{cl} to t_1 minus the return from t_s^{cl} to t_s . Our estimate of the bid-ask spread is the return from t_s to t_s^{cl} minus the return from t_b to t_b^{cl} . When the return from timing is added to the main calculation and the spread is subtracted, one gets the (approximate) return from t_b to t_s , the period in which the investor held the stock.

Table A1: The Gross Abnormal Returns for Stocks Bought and Sold in the Sample from the Day Following a Purchase or Sale to the End of the Month, January 1991 to November 1996

Purchase turnover is the average value of stocks purchased divided by the average value of stocks held in each month. The purchase abnormal return is calculated by compounding the daily returns on the purchased security from the day following the purchase to the end of the month less the compound return on the value-weighted NYSE/ASE/NASDAQ market index. Sales turnover and sales abnormal return are analogously calculated. The estimated effect on the monthly abnormal return is: purchase turnover times purchase abnormal return minus sale turnover times sale abnormal return.

Sample	Monthly Purchase Turnover (%)	Purchase Abnormal Return (%)	Monthly Sale Turnover (%)	Sale Abnormal Return (%)	Estimated Effect on Monthly Abnormal Return (%)
Panel A: Aggregate					
All Households	5.31%	-0.450%	4.93%	0.011%	-0.0245%
Panel B: Households partitioned by Beginning Position Value					
1 (Small)	7.08	-0.649	6.23	-0.148	-0.0367
2	6.39	-0.281	5.54	-0.060	-0.0147
3	6.03	-0.516	5.37	0.538	-0.0600
4	5.99	-0.381	5.56	-0.080	-0.0183
5 (Large)	4.70	-0.458	4.46	-0.026	-0.0204
Panel C: Households partitioned by Household Type					
General	4.76	-0.511	4.82	0.050	-0.0267
Affluent	2.06	-0.220	1.82	0.010	-0.0047
Active Trader	9.08	-0.475	8.14	-0.011	-0.0422
Panel D: Households partitioned by Turnover					
1 (Low)	0.20	-0.215	0.18	0.197	-0.0008
2	1.38	-0.142	1.12	-0.018	-0.0017
3	3.32	-0.106	3.01	0.060	-0.0053
4	7.08	-0.222	6.66	0.051	-0.0191
5 (High)	23.74	-0.654	22.37	-0.014	-0.1522

Appendix B: The Analysis of Intramonth Trades

In this appendix, we analyze the performance of stocks that are bought and then sold within a calendar month (e.g. purchased on January 3 and sold on January 10). These intramonth trades are excluded from our main analyses, since those analyses are based on monthly position statements. In aggregate, we estimate that intramonth trades would improve the performance of individual investors by less than one basis point per month (or approximately 0.10 per cent annually). Though profitable, the aggregate value of intramonth trades account for less than 1.5 per cent of the aggregate value of positions held.

For each account, we identify all purchases followed by a sale within the same month. In accounting for multiple purchases and sales, we assume that the first securities purchased are the first sold. Over our 71-month sample period, we identify 87,095 round-trip intramonth trades worth approximately \$27 million per month, on average. In contrast, the average beginning-of-month value of positions held, which we analyze in the main text, is over \$1.8 billion.

We calculate the gross returns on these round-trip transactions using the CRSP daily return files assuming the security is purchased and sold at the close of trading on the purchase and sale date, respectively. We calculate the net returns on these round-trip transactions by subtracting estimates of the bid-ask spread and commissions as is done in the main text for the case of monthly returns. The average round-trip trade involves a purchase of \$22,275, is held for 6.16 days, and costs 2.08 per cent in commissions and 0.30 per cent for the bid-ask spread. (In aggregate, these round-trip trades cost 0.87 per cent in commissions and 0.27 per cent for the bid-ask spread.) Note that the bid-ask spread is lower than that documented for trades that we analyze in the main text, which have an average round-trip bid-ask spread of one per cent (see Table 2). This lower spread is likely a result of the intraday return earned by investors from the transaction price through the end of the trading day (which is included in our estimate of the spread) rather than a smaller bid-ask spread for these intramonth trades.

In table B1, we summarize our analysis of the gross and net returns earned on intramonth trades. In this table, we calculate market-adjusted abnormal returns by subtracting the daily value-weighted NYSE/ASE/NASDAQ CRSP market index from the return earned on each intramonth trade. Both the gross and net abnormal returns in this table are weighted by the size of each trade, so that we can estimate the aggregate impact of these intramonth trades on the performance of individual investors.

Panel A presents results for all households. In aggregate, the intramonth trades earn impressive gross abnormal returns of 1.64 per cent. The net abnormal returns are 0.50 per cent. Since these intramonth trades average 1.49 per cent of the average value of positions held, we estimate that these intramonth trades would improve the performance of individual investors by 0.74 basis points per month (0.0050×0.0149) in aggregate. This small improvement in performance does not affect any of the conclusions that we present in the main text.

We also analyze the profitability of intramonth trades by partitioning households on the basis of account size (Panel B), household type (Panel C), and turnover (Panel D). In short, none of these results are so dramatic that they would lead us to qualify any of the results that we present in our main text. Those who benefit most from intramonth trades are those who trade most. Their intramonth trades improve their performance by 4.47 basis points per month (last row, last column, Panel D). Yet, we estimate that these investors underperform by a whopping 91 basis points per month (last row, table 10).

In conclusion, we should emphasize that the positive net returns earned on intramonth trades do not necessarily imply that individual investors have superior short-term trading ability. If investors have a disposition to sell winning investments and ride losing investments (as proposed by Shefrin and Statman (1985)), we would expect to observe positive abnormal returns on short-term round-trip trades.

Table B1: The Gross and Net Abnormal Returns earned on Intramonth Trades, January 1991 to November 1996

The gross abnormal return on intramonth trades is calculated as the compound return from the day following the purchase to the day of the sale less the compound return on a value-weighted NYSE/ASE/NASDAQ index. The net abnormal return is the gross abnormal return adjusted for the return earned on the day of the purchase or sale, the bid-ask spread, and the commission cost. The intramonth trades as a percentage of total position value are the average monthly value of intramonth purchases divided by the average monthly value of all stocks held. The estimated effect on monthly abnormal return is the net abnormal return times the intramonth trades as a percentage of total position value.

Sample	Mean Trade Size	Gross Abnormal Return (%)	Net Abnormal Return (%)	Intramonth Trades as a % of Total Position Value	Estimated Change in Monthly Abnormal Return (%)
Panel A: Aggregate					
All Households	\$22,275	1.636%	0.496%	1.49%	0.0074%
Panel B: Households partitioned by Beginning Position Value					
1 (Small)	17,467	2.407	0.936	2.21	0.0207
2	12,621	2.035	0.155	1.41	0.0022
3	17,105	1.760	0.506	1.78	0.0090
4	20,873	1.302	0.320	2.08	0.0066
5 (Large)	28,225	1.577	0.529	1.23	0.0065
Panel C: Households partitioned by Household Type					
General	15,806	1.823	0.259	0.92	0.0024
Affluent	20,021	2.606	1.195	0.14	0.0016
Active Trader	25,193	1.540	0.523	3.40	0.0178
Panel D: Households partitioned by Turnover					
1 (Low)	16,268	1.186	0.116	0.03	0.0000
2	13,719	2.779	0.756	0.05	0.0003
3	17,298	1.995	0.475	0.22	0.0010
4	16,778	2.378	0.973	0.78	0.0074
5 (High)	23,468	1.540	0.445	10.05	0.0447

Table 1

Summary of Sample Characteristics for Household Accounts
February 1991 - December 1996

The sample consists of 64,715 households with accounts at a discount brokerage firm.

Number of Households	64,715
Number of Accounts	128,848
Number of Trades in all Securities	3,075,797
Percentage of Trades in Common Stock	64.04%
Percentage of Account Value in Common Stocks	59.64%
Average Number of Common Stocks Held in Each Month	
Mean	4.03
25 th Percentile	1.35
Median	2.48
75 th Percentile	4.68
Std. Deviation	5.21
Average Monthly Value of Total Common Stock Holdings	
Mean	41,344
25 th Percentile	6,785
Median	14,579
75 th Percentile	33,324
Std. Deviation	177,917

Table 2

Descriptive Statistics on Trade Size, Trade Price, Transaction Costs, and Turnover:
February 1991 to December 1996

The sample is drawn from the account records for 64,715 households at a large discount brokerage firm. Spread is calculated as the transaction price divided by the closing price on the day of the transaction minus one (and then multiplied by minus one for purchases). Commission is calculated as the commission paid divided by the value of the trade. Monthly turnover is the beginning-of-month market value of shares purchased in month t-1 (or sold in month t) divided by the total beginning-of-month market value of shares held in month t. Trade-weighted spread and commission are averages weighted by trade size. Trade-weighted turnover is the aggregate value of sales (or purchases) divided by the aggregate value of positions held during our sample period.

	Mean	25 th Perc.	Median	75 th Perc.	Std. Dev.	Obs.
Panel A: Purchases						
Trade Size (\$)	11,205	2,513	4,988	10,500	32,179	1,082,107
Price/Share	31.06	11.00	23.00	40.00	117.82	1,082,107
Spread(%)*	0.31	-0.78	0.00	1.33	3.27	1,028,087
Commission (%)	2.09	0.82	1.40	2.41	12.38	1,076,565
Monthly Turnover (%)	7.28	0.34	2.95	8.07	13.34	64,715
Panel B: Sales						
Trade Size (\$)	13,707	2,688	5,738	13,000	38,275	887,594
Price/Share	31.22	12.00	24.00	41.00	113.03	887,594
Spread (%)*	0.69	-0.46	0.02	1.45	3.11	845,644
Commission (%)	3.07	0.73	1.25	2.24	140.50	881,761
Monthly Turnover (%)	6.58	0.02	2.48	7.39	12.23	64,715
Panel C: Trade-Weighted Purchases						
Spread (%)*	0.27					
Commission (%)	0.77				Not Applicable	
Monthly Turnover (%)	4.91					
Panel D: Trade-Weighted Sales						
Spread (%)*	0.61					
Commission (%)	0.66				Not Applicable	
Monthly Turnover (%)	4.47					

* Spread includes an intraday return on the day of the trade.

Table 3

**Summary of Percentage Monthly Abnormal Return Measures for the Average Household
and Aggregate Household: February 1991 to December 1996**

Panel A (Panel C) presents results for the gross (net) return on a portfolio that mimics the aggregate investment of all households. Panel B (Panel D) presents results for the gross (net) return on a portfolio that mimics the investment of the average household. Market-adjusted return is the return on the household less the return on a value-weighted NYSE/ASE/NASDAQ index. CAPM intercept is the estimated intercept from a time-series regression of the household excess return on the market excess return ($R_{mt} - R_{ft}$). Fama-French Three-Factor are the results from a time-series regressions of household excess return on the market excess return, a zero-investment book-to-market portfolio (HML_t), and a zero-investment size portfolio (SMB_t). Standard errors are presented in parentheses.

Coefficient Estimate on:					
	Excess Return	$(R_{mt} - R_{ft})$	HML_t	SMB_t	Adjusted R^2
Panel A: Gross Value-Weighted Percentage Monthly Returns					
Market Adjusted	0.016 (0.108)				
CAPM Intercept	-0.077 (0.110)	1.093** (0.036)			92.8
Fama-French Three-Factor	-0.095 (0.080)	1.077*** (0.028)	-0.023 (0.034)	0.239*** (0.030)	96.4
Panel B: Gross Average Household Percentage Monthly Returns					
Market Adjusted	0.064 (0.186)				
CAPM Intercept	-0.017 (0.197)	1.081 (0.065)			79.7
Fama-French Three-Factor	-0.168 (0.120)	1.111** (0.042)	0.146*** (0.051)	0.521*** (0.045)	93.0
Panel C: Net Value-Weighted Percentage Monthly Returns					
Market Adjusted	-0.097 (0.106)				
CAPM Intercept	-0.186* (0.109)	1.089** (0.036)			92.8
Fama-French Three-Factor	-0.201** (0.080)	1.071** (0.028)	-0.028 (0.034)	0.234*** (0.030)	96.4
Panel D: Net Average Household Percentage Monthly Returns					
Market Adjusted	-0.109 (0.183)				
CAPM Intercept	-0.184 (0.194)	1.075 (0.064)			80.0
Fama-French Three-Factor	-0.329*** (0.118)	1.103** (0.041)	0.138*** (0.050)	0.512*** (0.045)	93.1

***, **, * - significant at the 1, 5, and 10% level, respectively. The null hypothesis for beta (the coefficient estimate on the market excess return) is $H_0: \beta = 1$.

Table 4

Descriptive Statistics on Beginning Position Value, Portfolio Turnover, and Fama-French Three-Factor Loadings by Household Type: February 1991 to December 1996

The discount brokerage firm assigns account types. The firm labels households that make more than 48 trades in any year as active traders, other households with more than \$100,000 in equity at any time as affluent, and all other households as general. Beginning position value is the market value of common stocks held in the first month that the household appears during our sample period. Mean monthly turnover is the average of sales and purchase turnover. Coefficient estimates are those from a time-series regression of the gross average household excess return on the market excess return ($R_{mt} - R_{ft}$), a zero-investment book-to-market portfolio (HML_t), and a zero-investment size portfolio (SMB_t).

	General	Affluent	Active Trader
Number of Households	48,146	10,747	5,822
Mean Beginning Position Value	14,661	47,354	68,055
Mean Monthly Turnover (%)	6.74	4.60	12.78
Coefficient Estimate on: ($R_{mt} - R_{ft}$)	1.11***	1.07***	1.19***
HML_t	0.16***	0.12***	0.06
SMB_t	0.56***	0.29***	0.67***
Adjusted R ²	92.5	94.8	91.4

***, **, * - significant at the 1, 5, and 10% level, respectively.

Table 5

Percentage Monthly Returns for Portfolios formed by Household Type:
February 1991 to December 1996

The discount brokerage firm designates the household type. The firm labels households that make more than 48 trades in any year as active traders, other households with more than \$100,000 in equity at any time as affluent, and all other households as general. Raw return is the average monthly return for the average household. Market-adjusted return is the return on the household less the return on a value-weighted NYSE/ASE/NASDAQ index. CAPM intercept is the estimated intercept from a time-series regression of the household excess return on the market excess return ($R_{mt} - R_{ft}$). Fama-French Three-Factor are the results from a time-series regressions of household excess return on the market excess return, a zero-investment book-to-market portfolio (HML_t), and a zero-investment size portfolio (SMB_t).

	General	Affluent	Active Trader
Panel A: Gross Average Household Percentage Monthly Return			
Raw Return	1.386	1.516	1.537
Market-Adjusted Return	0.023	0.154	0.174
CAPM Intercept	-0.054	0.116	-0.018
Fama-French Intercept	-0.220*	0.012	-0.140
Panel B: Net Average Household Percentage Monthly Return			
Raw Return	1.218	1.420	1.191
Market-Adjusted Return	-0.145	0.056	-0.172
CAPM Intercept	-0.217	0.024	-0.353
Fama-French Intercept	-0.376***	-0.077	-0.468***

***, **, * - significant at the 1, 5, and 10% level, respectively.

Table 6

Descriptive Statistics on Beginning Position Value, Portfolio Turnover, and Fama-French Three-Factor Loadings for Quintiles formed on Beginning Position Value:
February 1991 to December 1996

Households are sorted into quintiles based on the market value of common stocks in the first month that a household appears during our sample period. Quintile one contains households with the smallest market value of common stock holdings, while quintile five contains households with the largest. Beginning position value is the market value of common stocks held in the first month that the household appears during our sample period. Mean monthly turnover is the average of sales and purchase turnover. Coefficient estimates are those from a time-series regression of the gross average household excess return on the market excess return ($R_{mt} - R_{ft}$), a zero-investment book-to-market portfolio (HML_t), and a zero-investment size portfolio (SMB_t).

Quintile:	1 (Small)	2	3	4	5 (Large)
Mean Beginning Position Value	1,599	4,700	19,161	30,386	133,262
Mean Monthly Turnover (%)	7.44	6.93	6.91	6.64	6.72
Coefficient Estimate on: ($R_{mt} - R_{ft}$)	1.20***	1.11***	1.10***	1.10***	1.05***
HML_t	0.37***	0.12**	0.10*	0.10**	0.07*
SMB_t	0.98***	0.56***	0.46***	0.39***	0.27***
Adjusted R ²	86.3	92.8	93.0	94.2	95.9

***, **, * - significant at the 1, 5, and 10% level, respectively.

Table 7

Percentage Monthly Returns for Quintile Portfolios
formed on Beginning Position Value: February 1991 to December 1996

Households are sorted into quintiles based on the market value of common stocks in the first month that a household appears during our sample period. Quintile one contains households with the smallest market value of common stock holdings, while quintile five contains households with the largest. Raw return is the average monthly return for the average household. Market-adjusted return is the return on the household less the return on a value-weighted NYSE/ASE/NASDAQ index. CAPM intercept is the estimated intercept from a time-series regression of the household excess return on the market excess return ($R_{mt} - R_{ft}$). Fama-French Three-Factor are the results from a time-series regressions of household excess return on the market excess return, a zero-investment book-to-market portfolio (HML_t), and a zero-investment size portfolio (SMB_t).

Quintile:	1 (Small)	2	3	4	5 (Large)	Difference Sml-Lrg
Panel A: Gross Average Household Percentage Monthly Return						
Raw Return	1.648	1.431	1.401	1.357	1.331	0.317
Market-Adjusted Return	0.285	0.068	0.038	-0.006	-0.032	0.317
CAPM Intercept	0.175	-0.025	-0.046	-0.083	-0.079	0.254
Fama-French Intercept	-0.162	-0.168	-0.162	-0.188*	-0.154**	0.008
Panel B: Net Average Household Percentage Monthly Return						
Raw Return	1.398	1.245	1.237	1.212	1.202	-0.196
Market-Adjusted Return	0.035	-0.118	-0.126	-0.151	-0.161	-0.196
CAPM Intercept	-0.069	-0.204	-0.205	-0.224	-0.203*	-0.134
Fama-French Intercept	-0.398**	-0.340***	-0.315***	-0.322***	-0.273***	0.125

***, **, * - significant at the 1, 5, and 10% level, respectively .

Table 8
Cross-Sectional Distribution of
Percentage Monthly Gross and Net Market-Adjusted Household Returns:
February 1991 to December 1996

Households with position statements in 12 or fewer months are omitted from this analysis. Though the median values are virtually identical when these households are included, more extreme values are observed.

	Gross Monthly Market-Adjusted Return (%)	Net Monthly Market-Adjusted Return (%)
Minimum	-19.46	-23.03
1 st Percentile	-4.46	-5.03
5 th Percentile	-2.32	-2.64
10 th Percentile	-1.49	-1.75
25 th Percentile	-0.60	-0.77
Median	0.04	-0.09
75 th Percentile	0.81	0.63
90 th Percentile	1.94	1.70
95 th Percentile	2.96	2.69
99 th Percentile	5.98	5.52
Maximum	46.19	45.48
Total Households	59,534	59,534
Percentage > 0	51.8%***	46.0%***
Binomial Z-statistic	8.90	-19.43

*** - significantly different from 50% at the 1% level.

Table 9

Descriptive Statistics on Beginning Position Value, Portfolio Turnover, and Fama-French Three-Factor Loadings for Quintiles formed on Mean Turnover:
February 1991 to December 1996

Households are sorted into quintiles based on monthly turnover (the average of sales and purchase turnover) during our sample period. Quintile one contains households with the lowest turnover, while quintile five contains households with the highest. Beginning position value is the market value of common stocks held in the first month that the household appears during our sample period. Mean monthly turnover is the average of sales and purchase turnover. Coefficient estimates are those from a time-series regression of the gross average household excess return on the market excess return ($R_{mt} - R_{ft}$), (HML_t), and a zero-investment size portfolio (SMB_t).

Quintile:	1 (Low)	2	3	4	5 (High)
Mean Monthly Turnover (%)	0.12	1.26	3.10	6.54	23.62
Mean Beginning Position Value	32,058	50,988	40,602	19,182	20,835
Coefficient Estimate on: ($R_{mt} - R_{ft}$)	1.03***	1.05***	1.10***	1.17***	1.28***
HML_t	0.22***	0.11***	0.12**	0.13*	0.12
SMB_t	0.27***	0.29***	0.50***	0.74***	1.05***
Adjusted R ²	95.4	95.0	92.2	90.4	87.7

***, **, * - significant at the 1, 5, and 10% level, respectively.

Table 10

Percentage Monthly Returns for Quintile Portfolios formed on Mean Turnover:
February 1991 to December 1996

Households are sorted into quintiles based on monthly turnover (the average of sales and purchase turnover) during our sample period. Quintile one contains households with the lowest turnover, while quintile five contains households with the highest. Raw return is the average monthly return for the average household. Market-adjusted return is the return on the household less the return on a value-weighted NYSE/ASE/NASDAQ index. CAPM intercept is the estimated intercept from a time-series regression of the household excess return on the market excess return ($R_{mt} - R_{ft}$). Fama-French Three-Factor are the results from a time-series regressions of household excess return on the market excess return, a zero-investment book-to-market portfolio (HML_t), and a zero-investment size portfolio (SMB_t).

Quintile:	1 (Low)	2	3	4	5 (High)	Difference Low-High
Panel A: Gross Average Household Percentage Monthly Return						
Raw Return	1.408	1.424	1.407	1.422	1.478	0.070
Market-Adjusted Return	0.045	0.061	0.044	0.059	0.115	0.070
CAPM Intercept	0.077	0.038	-0.032	-0.097	-0.165	-0.242
Fama-French Intercept	-0.084	-0.061	-0.165	-0.268*	-0.372*	-0.288
Panel B: Net Average Household Percentage Monthly Return						
Raw Return	1.398	1.362	1.274	1.165	0.907	-0.491*
Market-Adjusted Return	0.035	-0.001	-0.089	-0.198	-0.456	-0.491*
CAPM Intercept	0.067	-0.022	-0.161	-0.344	-0.719*	-0.787**
Fama-French Intercept	-0.094	-0.119	-0.291**	-0.506***	-0.906***	-0.811***

***, **, * - significant at the 1, 5, and 10% level, respectively.

Table 11

Summary of Percentage Monthly Abnormal Return Measures for Vanguard Index 500:
February 1991 to December 1996

Market-adjusted return is the return on the Vanguard Index 500 less the return on a value-weighted NYSE/ASE/NASDAQ index. CAPM intercept is the estimated intercept from a time-series regression of the Vanguard Index 500 excess return on the market excess return ($R_{mt} - R_{ft}$). Fama-French Three-Factor are the results from a time-series regressions of Vanguard Index 500 excess return on the market excess return, a zero-investment book-to-market portfolio (HML_t), and a zero-investment size portfolio (SMB_t). Standard errors are in parentheses.

	Coefficient Estimate on:				Adjusted R^2
	Excess Return	$(R_{mt} - R_{ft})$	HML_t	SMB_t	
Market Adjusted	-0.013 (0.065)				
CAPM Intercept	-0.009 (0.069)	0.996*** (0.023)			96.4
Fama-French Three-Factor	0.003 (0.034)	1.010*** (0.012)	0.021 (0.014)	-0.183*** (0.013)	99.2

*** - significantly different from zero at the 1% significance level.

Table 12

Descriptive Statistics on Beginning Position Value, Portfolio Turnover, and Fama-French Three-Factor Loadings for Taxable and Tax-Deferred Accounts:
February 1991 to December 1996

Accounts are partitioned as either taxable or tax deferred (IRA, Keogh, SEP-IRA). Beginning position value is the market value of common stocks held in the first month that the household appears during our sample period. Mean monthly turnover is the average of sales and purchase turnover. Coefficient estimates are those from a time-series regression of the gross average household excess return on the market excess return ($R_{mt} - R_{ft}$), a zero-investment book-to-market portfolio (HML_t), and a zero-investment size portfolio (SMB_t).

	Taxable	Tax-Deferred	Difference
Number of Households	52,338	28,296	N/A.
Mean Beginning Position Value	26,379	13,545	12,834***
Mean Monthly Turnover (%)	8.07	6.04	2.03***
Coefficient Estimate on: $(R_{mt} - R_{ft})$	1.12***	1.12***	0.00
HML_t	0.15***	0.19***	-0.04***
SMB_t	0.56***	0.53***	0.03***
Adjusted R ²	92.5	92.3	36.3

***, **, * - significant at the 1, 5, and 10% level, respectively.

Table 13

Percentage Monthly Returns for Taxable and Tax-Deferred Portfolios:
February 1991 to December 1996

Accounts are partitioned as either taxable or tax deferred (IRA, Keogh, SEP-IRA). Raw return is the average monthly return for the average household. Market-adjusted return is the return on the household less the return on a value-weighted NYSE/ASE/NASDAQ index. CAPM intercept is the estimated intercept from a time-series regression of the household excess return on the market excess return ($R_{mt} - R_{ft}$). Fama-French Three-Factor are the results from a time-series regressions of household excess return on the market excess return, a zero-investment book-to-market portfolio (HML_t), and a zero-investment size portfolio (SMB_t).

	Taxable	Tax-Deferred	Difference
Panel A: Gross Average Household Percentage Monthly Return			
Raw Return	1.426	1.462	-0.036
Market-Adjusted Return	0.063	0.099	-0.036
CAPM Intercept	-0.028	0.026	-0.054*
Fama-French Intercept	-0.185	-0.149	-0.036
Panel B: Net Average Household Percentage Monthly Return			
Raw Return	1.239	1.303	-0.064**
Market-Adjusted Return	-0.124	-0.060	-0.064**
CAPM Intercept	-0.208	-0.129	-0.079***
Fama-French Intercept	-0.359***	-0.298**	-0.061**

***, **, * - significant at the 1, 5, and 10% level, respectively.

**Figure 1: Cross-Sectional Distribution of
Percentage Mean Gross Monthly Market-Adjusted Abnormal Return: 2/91 - 12/96**

